MCP Quiz (Model Context Protocol)

◆ Section A: MCQs (1 Mark Each)

1. What is MCP primarily used for?
A. Data visualization
B. Agent-to-tool communication
C. Database management
D. UI design

Answer: B. Agent-to-tool communication

---

2. MCP stands for:
A. Model Control Pipeline
B. Machine Context Processing
C. Model Context Protocol
D. Managed Compute Platform

Answer: C. Model Context Protocol

---

3. In MCP architecture, what is the role of an "Agent"?
A. Stores data
B. Executes database queries
C. Decides actions and interacts with tools
D. Designs UI

Answer: C. Decides actions and interacts with tools

---

4. Which component actually performs tasks like API calls?
A. Agent
B. Tool
C. Context
D. Connector

Answer: B. Tool

---

5. MCP helps in:
A. Replacing APIs
B. Standardizing communication between AI and tools
C. Removing cloud dependency
D. Building databases

Answer: B. Standardizing communication between AI and tools

◆ Section B: True / False (1 Mark Each)

6. MCP eliminates the need for tools.

Answer: False

7. Agents can call multiple tools using MCP.

Answer: True

8. MCP only works with Python.

Answer: False

9. Context in MCP helps the agent make better decisions.

Answer: True

10. MCP architecture includes Agent, Tool, and Context.

Answer: True

◆ Section C: Short Answer (2 Marks Each)

11. Define Model Context Protocol (MCP).

Answer: Model Context Protocol (MCP) is a structured framework that standardizes how an AI agent communicates with external tools, APIs, and data sources by passing the right context for task execution.

---

12. What is “context” in MCP?

Answer: Context is the relevant information provided to the agent or tool, such as user input, memory, previous conversation, system state, or task details, so the agent can make accurate decisions.

---

13. Give two examples of MCP tools.

Answer: Two examples of MCP tools are a weather API tool and a database query tool. Other valid examples include calculator tools, search tools, or file-reading tools.

---

14. What is the difference between an Agent and a Tool in MCP?

Answer: The agent is the decision-maker that understands the request, plans the next step, and decides which tool to use. A tool is the executor that performs the actual task, such as calling an API, searching a database, or doing a calculation.

---

15. Why are connectors important in MCP?

Answer: Connectors are important because they link the agent to external systems, services, or data sources. They make integration smoother and allow the agent to interact with real-world tools in a standardized way.

◆ Section D: Scenario-Based Questions (3 Marks Each)

16. Scenario:
An AI assistant receives a user query: *"What is the weather in Delhi?"*

Answer:

- Agent role: The agent understands the user query, decides that weather information is needed, and selects the correct tool.

- Tool used: A weather API tool.

- How MCP enables it: MCP provides a standard way for the agent to pass the city name, request current weather data, receive the result, and return a useful response to the user.

- Identify the role of the agent
- Identify the tool used
- Explain how MCP enables this interaction

---

17. Scenario:
A chatbot needs to fetch customer data from a database and then calculate a discount.

Answer:

- Tools involved: Two tools can be involved - one database tool to fetch customer data and one calculation or business-logic tool to compute the discount.

- Role of MCP: MCP coordinates the communication between the agent and both tools, ensuring the right data and context move step by step from fetching data to calculating the discount.

- How many tools are involved?
- What is the role of MCP here?

---

18. Scenario:
An agent fails to provide accurate output due to missing context.

Answer:

- What went wrong: The agent did not receive enough relevant context, so it made a weak or incorrect decision.

- How MCP can fix it: MCP can supply structured context such as user history, task details, system rules, or previous tool outputs, helping the agent produce a more accurate result.

- What went wrong?
- How can MCP fix this issue?

◆ Section E: Advanced / Conceptual (5 Marks Each)

19. Explain MCP architecture with its key components.

Answer: MCP architecture mainly consists of three key components: Agent, Tool, and Context. The agent is the intelligent decision-maker that interprets the user request and decides what action to take. Tools are the functional units that execute tasks such as API calls, calculations, file access, or database queries. Context is the supporting information that helps the agent choose correctly and respond accurately. In many setups, connectors are also used to bridge tools with external systems. Together, these parts create a standardized, modular, and scalable way for AI systems to interact with real-world services.

---

20. Describe how Agent–Tool communication works in MCP step by step.

Answer: Step 1: The user gives a query to the agent. Step 2: The agent analyzes the request and identifies the required task. Step 3: The agent selects the appropriate tool. Step 4: MCP formats and passes the necessary context and parameters to that tool. Step 5: The tool performs the task, such as calling an API or fetching data. Step 6: The tool sends the result back through the MCP structure. Step 7: The agent interprets the output and presents the final answer to the user. This step-by-step flow makes communication clear, reusable, and reliable.